SESSION 3

FINANCIAL RATIO ANALYSIS

SCALER SCHOOL OF TECHNOLOGY

Learning Outcomes

Session 3: Financial Ratio Analysis

By the end of the session, learners will be able to:

- **Explain** the purpose and importance of financial ratio analysis.
- **Describe** key profitability, return, efficiency, working capital, liquidity, and leverage ratios.
- **Interpret** these ratios to assess a company's performance, financial health, efficiency, returns, and risk.
- Apply financial ratios to support business evaluation and decision-making.

Financial Ratios: Introduction

Financial ratios are critical metrics used to **evaluate** a **company's financial health** and **performance.**

✔ It is derived from a firm's financial statements.

✔ These ratios facilitate comparisons both within the company over time and against industry peers.

Objectives of financial ratios:

- ❑ To **assess profitability** – whether the business generates **sufficient profits** for its owners and other stakeholders
- ❑ To **evaluate liquidity** – whether the company can meet its **short-term obligations.**
- ❑ To **measure operational efficiency** – whether the company uses its assets efficiently
- ❑ To **assess solvency** and **financial risk** – whether the company can meet its **long-term obligations**
- ❑ To **evaluate returns on capital employed** – provided by **owners and lenders**
- ❑ To facilitate **performance comparisons** – across **time, competitors, industry & budgeted figures**
- ❑ To support **managerial decision-making**

Financial Ratios: Types

Financial ratios can be grouped into five major categories:

- **Liquidity Ratios –** Can the business meet its **short-term obligations?**
- **Efficiency Ratios –** How **efficiently** is the business using its **assets** and working capital?
- **Solvency/Leverage Ratios –** Can the business meet its **long-term obligations**, and how much financial risk does it carry?
- **Profitability & Return Ratios –** How effectively is the **business generating profit** and returns?
- **Market/Investor Ratios –** How is the **business performing** from the perspective of shareholders and the stock market?

Financial Ratios: Types

Liquidity Ratios: They are financial ratios used to measure a company's **ability to meet its short-term financial obligations** as and when they become due.

"Does the business have enough short-term assets and cash-generating capacity to pay its short-term liabilities?"

Short-term liabilities generally include obligations that are payable **within one year,** such as:

- Trade payables/creditors
- Short-term borrowings
- Bank overdrafts
- Accrued expenses
- Outstanding wages and salaries
- Taxes payable
- Current portion of long-term debt

The main short-term assets available to meet these obligations include:

- Cash and cash equivalents
- Bank balances
- Short-term investments/marketable securities
- Trade receivables
- Inventory
- Other current assets

Financial Ratios: Types

Types of Liquidity Ratios:

- **Current Ratio:** It measures the **relationship** between a company's **current assets & current liabilities**.
  - ✔ It indicates the extent to which current assets are available to cover current liabilities.

Current Ratio is calculated as follows:

$$\text{Current Ratio} = \frac{\text{Current Assets}}{\text{Current Liabilities}}$$

Current assets = Cash & bank balance, inventories, trade receivables, prepaid expenses, short-term investments.

Current liabilities = Trade payables, bank overdraft, short-term borrowings, accrued expenses & other short-term obligations

Financial Ratios: Types

Types of Liquidity Ratios:

- **Quick Ratio**: Also known as the **Acid-Test Ratio,** it measures whether a company can meet its current liabilities using its **most liquid current assets,** without relying heavily on the sale of inventory.
  - ✔ This is important because **inventory** may take time to sell and **may not always be converted into cash easily.**

Quick Ratio is calculated as follows:

$$\text{Quick Ratio} = \frac{\text{Quick Assets}}{\text{Current Liabilities}}$$

Quick assets = Current Assets - Inventory

- **Cash Ratio**: It only considers **cash & cash equivalents** and other highly liquid marketable securities.

  Cash Ratio is calculated as follows:

$$\text{Cash Ratio} = \frac{\text{Cash} + \text{Cash Equivalents} + \text{Marketable Securities}}{\text{Current Liabilities}}$$

Financial Ratios: Types

Efficiency Ratios: They are also called **activity ratios or asset-management ratios,** measure how **effectively** a business **uses its assets** and **working capital** to generate sales and revenue.

"How efficiently is the company using the resources available to it to generate revenue?"

A business invests money in:

- Inventory
- Trade receivables
- Working capital
- Fixed assets
- Total assets

Efficiency ratios assess whether these resources are being used productively.

Financial Ratios: Types

Types of Efficiency Ratios:

- **Inventory Turnover Ratio:** It measures **how many times** a company sells and replaces its inventory during a particular accounting period.
- Inventory Turnover Ratio is calculated as follows:

$$\text{Inventory Turnover Ratio} = \frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Where,

$$\text{Average Inventory} = \frac{\text{Opening Inventory} + \text{Closing Inventory}}{2}$$

- **Inventory Holding Period:** Also called **Days Inventory Outstanding,** it measures the **average number of days inventory remains in the business** before being sold
- It is calculated as:

$$\text{Inventory Holding Period} = \frac{\text{Average Inventory}}{\text{COGS}} \times 365$$

Financial Ratios: Types

Types of Efficiency Ratios:

- **Receivables Turnover Ratio:** It measures **how efficiently** a company manages its trade receivables and **collects money from customers.**
- It indicates **how many times**, on average, the company **converts its receivables into cash** during the year.
- It is calculated as:

$$\text{Receivables Turnover Ratio} = \frac{\text{Net Credit Sales}}{\text{Average Trade Receivables}}$$

where,

$$Average\ Trade\ Receivables = \frac{Opening\ Receivables + Closing\ Receivables}{2}$$

- **Average Collection Period:** Also called as **Days Sales Outstanding** or **Debtor Days,** it measures **the average number of days customers take to pay the company.**
- It is calculated as:

$$\text{Average Collection Period} = \frac{\text{Average Trade Receivables}}{\text{Net Credit Sales}} \times 365$$

Financial Ratios: Types

Types of Efficiency Ratios:

- **Payables Turnover Ratio:** It measures **how efficiently** a company pays its suppliers during a given period.
- It is calculated as: $$\text{Payables Turnover Ratio} = \frac{\text{Credit Purchases}}{\text{Average Trade Payables}}$$

where, $$Average\ Trade\ Payables = \frac{Opening\ Payables + Closing\ Payables}{2}$$

If credit purchases are unavailable, COGS can be used instead.

- **Average Payment Period:** Also called as **Days Payable Outstanding** or **Payables Days,** it measures **the average number of**
- It is calculated as: $$\text{Average Payment Period} = \frac{\text{Average Trade Payables}}{\text{Credit Purchases}} \times 365$$

Financial Ratios: Types

Working Capital Cycle:

- The **Working Capital Cycle,** also called the **Operating Cycle,** measures **the time taken by a business to convert the money invested in its day-to-day operations back into cash.**

It is calculated as follows:

Operating Cycle = Inventory Holding Period + Receivable Days – Payable Days

Financial Ratios: Types

Solvency Ratios: Also called **leverage ratios,** it measures a **company's ability to meet its long-term financial obligations.**

“Does the company have a financially sustainable level of debt, and can it meet its long-term interest and principal repayment obligations?”

While **liquidity ratios** focus mainly on the **short term,** **solvency ratios** focus on the company's **long-term financial stability** and **capital structure.**

Financial Ratios: Types

Types of Solvency Ratios:

- **Debt-to-Equity (D/E) Ratio:** It compares the **amount of debt financing** with **shareholders' equity.**
- It helps determine how much the company relies on **borrowed funds** relative to **shareholders' funds.**
- It is calculated as: $$Debt\text{-}to\text{-}Equity = \frac{Total\ Debt}{Shareholders'\ Equity}$$

Where, Debt = total interest-bearing debt or total liabilities

Equity = Share capital + Reserves & Surplus (retained earnings)

Financial Ratios: Types

Types of Solvency Ratios:

- **Interest Coverage Ratio:** It measures the **company's ability to pay its interest expenses** from its **operating profits.**

  "How many times can the company's operating profit cover its interest expense?"

- It is calculated as:

$$Interest\ Coverage\ Ratio = \frac{EBIT}{Interest\ Expense}$$

Where, EBIT = Earnings before interest & tax

Thank You!